

Module 1: AI Basics - Complete Guide

This guide covers everything you need to know about Artificial Intelligence fundamentals.

1.1 What is Artificial Intelligence?

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think and learn like humans.

- AI is the ability of a machine to perform tasks that typically require human intelligence
- Includes learning, reasoning, problem-solving, perception, and language understanding
- AI can be narrow (specific tasks) or general (any intellectual task)

1.2 History of AI

The field of AI research was founded at a workshop at Dartmouth College in 1956.

- 1950s: Alan Turing proposes the Turing Test
- 1956: Term "Artificial Intelligence" coined at Dartmouth Conference
- 1980s-90s: Expert systems and neural networks research
- 2010s: Deep learning revolution begins
- 2020s: Large language models and generative AI explode

1.3 Machine Learning vs AI

Machine Learning (ML) is a subset of AI.

- AI: The broad concept of machines mimicking human intelligence
- ML: A subset of AI where machines learn from data
- Deep Learning: A subset of ML using neural networks with many layers

1.4 Key AI Concepts

- Neural Networks: Computing systems inspired by biological neural networks
- Natural Language Processing (NLP): AI ability to understand human language
- Computer Vision: AI ability to interpret visual information
- Robotics: AI-powered machines that can interact with the physical world

- Generative AI: AI that can create new content (text, images, video, code)

1.5 AI Applications Today

- Healthcare: Disease diagnosis, drug discovery, personalized medicine
- Finance: Fraud detection, algorithmic trading, risk assessment
- Transportation: Autonomous vehicles, route optimization
- Education: Personalized learning, automated grading
- Business: Customer service chatbots, data analysis, automation
- Creative: Image generation, music composition, content writing